



- 9 (a) A stack has been implemented using pseudocode to store a maximum of 100 string items using the global variables in the following table:

Identifier	Data type	Description	Initialisation value
Base	INTEGER	pointer for the bottom of the stack	0
Top	INTEGER	pointer for the top of the stack	-1
StackArray	STRING	1D array to implement the stack	[0:99]
Max	INTEGER	maximum number of items in the stack	100

The value of `Top` is incremented each time a data item is added to the stack and decremented every time a data item is removed.

- (i) Complete the **pseudocode** for the function to remove a data item from the stack.

```

FUNCTION Pop() .....
    DECLARE DataItem : STRING
    DataItem ← ""

    IF ..... THEN

        DataItem ← .....

        Top ← .....
    ELSE
        DataItem ← "You cannot remove data; the stack is empty"
    ENDIF

    .....
ENDFUNCTION
    
```

[5]

- (ii) Write the **pseudocode** to output the data item removed from the stack with an appropriate message.

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..... [1]

